Deep Learning

Assignment 12

Discussion on: 11.02.2026

Question 1 Computational costs of convolutions (10 points)

Suppose in a neural network, we have two tensors A and B with sizes $3 \times 32 \times 32$ and $16 \times 32 \times 32$ (channel, height, width). Include the necessary padding, ignore the bias tensor and batch size.

(a) Compute the number of multiplication operations if we convolve a kernel of spatial size 5×5 so that $B = \mathrm{conv}(A)$ (with a stride of 1). (3 points)

(b) Compute the number of multiplication operations if we convolve two kernels each of spatial size 3×3 so that $H = \mathrm{conv}_1(A)$, $B = \mathrm{conv}_2(H)$ where H has n channels. (2 points)

(c) Choose n as large as possible, but so that (b) requires less multiplication operations than (a). (1 point)

(d) Regular convolutions might have redundancy in their weights, so it is often more efficient to factorize spatial and cross-channel into two steps (*depthwise separable convolutions*). For this, we first convolve the kernel of (a) without any cross-channel computations. That means, that each channel is convolved with an individual kernel, resulting in one output channel per input channel. This leads to a tensor of size $3 \times 32 \times 32$. Afterwards we apply a regular kernel of spatial size 1×1. The final output results in B.
$H = \mathrm{conv}_{spatial}(A)$, $B = \mathrm{conv}_{channels}(H)$ (3 points)

(e) Compare the numbers of multiplication operations of (a) and (d), comment on the result. (1 point)

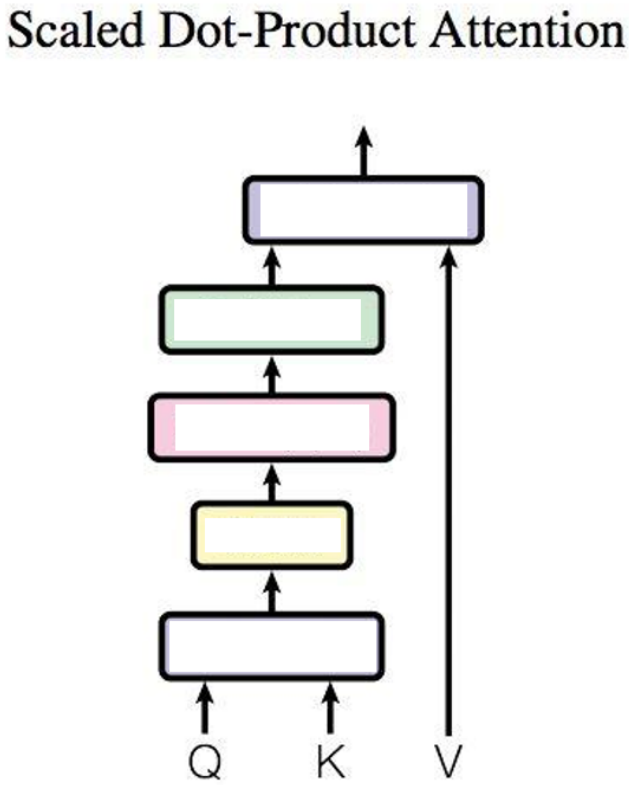
Question 2 Recurrent Neural Networks and Transformer (10 Points)

(a) What is the key idea of *self-attention* in a Transformer, compared to recurrent networks? (1 point)

(b) Given the following equations, draw the computational graph to compute the training loss of the corresponding recurrent neural network and unfold it for the time steps $t-1$, t, and $t+1$. (4 points)

$$
\begin{aligned}
\mathbf{a}^{(t)} &= \mathbf{b} + \mathbf{W}\mathbf{h}^{(t-1)} + \mathbf{U}\mathbf{x}^{(t)} \\
\mathbf{h}^{(t)} &= \tanh\left(\mathbf{a}^{(t)}\right) \\
\mathbf{o}^{(t)} &= \mathbf{c} + \mathbf{V}\mathbf{h}^{(t)} \\
\hat{\mathbf{y}}^{(t)} &= \mathrm{softmax}(\mathbf{o}^{(t)})
\end{aligned}
$$

(c) Write down the corresponding names next of the five white nodes and complete the equation for the self-attention (5 points)



$$\mathbf{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$